



KLE Technological
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**Department
of
Electronics and Communication Engineering**

**Mini Project Report
on
9V USB MOBILE CHARGER**

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
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CERTIFICATE

This is to certify that project entitled “**9v USB MOBILE CHARGER**” is a Bonafide work carried out by the student team of “**Kavana Gunaga(01fe23bec263), Vrunda Patil(01fe23bec248), Spoorti B.(01fe23bec246), Shreyas Wali(01fe23bec249)**”. The project report has been approved as it satisfies the requirements with respect to the Mini project work pre-scribed by the university curriculum for BE (V semester) in Department of Electronics and Communication Engineering of KLE Technological University for the academic year 2025-26.

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-Team E3

ABSTRACT

A regulated power supply is a fundamental requirement for the safe operation of portable electronic devices. Mobile phones and similar equipment demand a stable DC voltage for efficient charging, protection of internal circuits, and long battery life. Variations in supply or excessive ripple can lead to overheating, improper charging behaviour, and potential damage. Therefore, the design of a reliable and economical charger becomes an important engineering task.

This project presents the design and implementation of a **9 V, 1 A regulated mobile charger** using a step-down transformer, bridge rectifier, filter capacitor, and the 7809 linear voltage regulator. The AC mains input is first reduced to a suitable level, converted into pulsating DC, and then smoothed using an electrolytic capacitor. The regulator stage maintains a constant output voltage despite fluctuations in input or load conditions. To improve operational safety, a series fuse is incorporated at the output to protect the connected device during overcurrent or short-circuit situations.

The proposed system was verified through simulation as well as hardware realization. Waveform analysis confirmed ripple reduction after filtering and stable voltage regulation at the output. The practical utility of the charger was demonstrated by successfully charging a mobile phone, validating the effectiveness and reliability of the design. The project provides valuable insight into AC–DC conversion, regulation techniques, protection mechanisms, and real-world implementation of power electronic systems.

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Chapter 1 Introduction

The rapid advancement in portable electronics has increased the need for efficient and reliable charging solutions. Most electronic devices operate on DC power, while the available mains supply is AC. Therefore, conversion and regulation are necessary before the power can be safely used.

A charger must not only convert AC to DC but also ensure that the output remains constant under varying input voltage and load conditions. An unregulated supply may contain ripples and spikes that can adversely affect the performance of sensitive electronic circuits. This mini project focuses on developing a simple yet effective regulated charger using widely available components, incorporates fuse protection for short-circuit safety, and was practically verified by charging a mobile phone. Similar power supply designs for portable applications, such as a regulated 5 V, 1 A AC-to-DC supply using transformer, rectifier, filter, and regulator stages, have been studied in prior research to highlight component selection and design procedures for stable low-voltage DC sources[1]. The design illustrates the practical application of rectifiers, filters, voltage regulators, and protective elements, which are essential topics in electronics engineering.

1.1 Motivation

The motivation behind this project arises from the increasing dependency on mobile devices and the frequent need for safe charging methods. Commercial chargers often integrate complex circuits, making it difficult for students to understand their internal operation.

By constructing a charger using basic discrete components, we aim to gain a clear understanding of each stage involved in power conversion. This hands-on approach enhances practical knowledge and strengthens theoretical concepts learned in classrooms.

1.2 Objectives

The objectives of this mini project extend beyond simply producing a 9 V output. They include academic, practical, and analytical goals. The main objectives are:

- To design a safe and reliable AC to DC conversion unit.
- To implement voltage regulation using IC 7809.
- To ensure stable operation under varying load conditions.
- To compare simulation predictions with hardware observations.
- To validate the practical usability of the charger by successfully charging a mobile phone.
- To enhance user safety by incorporating fuse-based protection against short-circuit and overcurrent conditions.

1.3 Literature Survey

The design of regulated DC power supplies for portable electronic devices has been the subject of numerous research efforts. These studies provide valuable insights into the selection of components, topologies, protection mechanisms, and practical implementation strategies that improve reliability, efficiency, and safety of low-voltage charger circuits.

In [2], the authors present a detailed exploration of regulated DC power supply design targeting a **5 V, 1 A output**, using a step-down transformer, bridge rectifier, filter capacitor, and linear voltage regulator. The study emphasizes the importance of accurate calculation of ripple reduction and proper heat management for the linear regulator stage. Through simulation and hardware tests, the research illustrates how component tolerances and load conditions influence final output stability, thereby offering a foundational methodology applicable to other DC converter designs like the 9 V charger developed in this project.

Further expanding on practical implementation, ElectroThinks [3] discusses a **9 V, 1 A power supply circuit using a 7809 voltage regulator**, similar to the regulator used in the present work. This online reference provides schematic diagrams, component values, and practical objectives that align closely with our design. The discussion includes the rationale for transformer sizing, bridge rectifier arrangement, and placement of filter capacitors to minimize ripple. Such case studies serve as valuable benchmarks when validating real-world behavior and selecting appropriate part ratings.

The work shared on Scribd [4] describes a mobile charging application using the **7805 IC** with fundamental design stages, including rectification and regulation. While the target voltage is 5 V instead of 9 V, the design principles remain consistent, especially regarding the need for smoothing capacitors, regulator input headroom, and safe assembly practices. The documented implementation highlights common challenges students face during prototype construction — including thermal issues, incorrect polarities, and the practical effect of load changes — which informed our approach when incorporating protective elements such as the output fuse for our charger.

Simulation-based studies, such as that discussed on IvyPanda [5], provide an academic perspective on how transitional stages in power supplies influence output quality. This research emphasizes waveform analysis after each major stage — transformer output, rectifier conversion, filtering, and regulation — with simulated oscilloscope plots demonstrating ripple reduction and voltage stability. These insights were particularly useful during the simulation validation phase of our project, where observing waveform behavior helped identify optimal capacitor values and confirm that the 7809 regulator maintained regulation under different loads.

Lastly, research documented in [6] explores broader aspects of regulated AC-DC power supplies, including reliability evaluation and performance under varied load conditions. The paper delves into the comparative effectiveness of linear regulators versus other regulation methods and underscores the need for safety mechanisms such as fuses and transient protection. This aligns directly with the design decision in our prototype to include a fuse at the output for short-circuit protection, adding an important layer of device safety when charging a mobile phone.

In summary, these studies collectively support the circuit design and validation steps undertaken in the project. They attest to the importance of systematic design, practical simulations, and safety-focused enhancements in the development of regulated DC power supplies. By correlating theoretical research with hands-on validation, the current work integrates best practices and academic findings, reinforcing the functionality and reliability of the developed 9 V, 1 A charger.

1.4 Market Survey

The market for mobile charging solutions has expanded rapidly due to the widespread adoption of smartphones, tablets, and portable electronic devices. Leading manufacturers such as Apple, Samsung, Xiaomi, Oppo, Vivo, Realme, Anker, and Samsung Electronics design chargers that deliver stable output voltage, fast-charging capability, and multiple layers of safety protection, while maintaining compact size and affordability. Studying these commercial products helps in understanding current industry standards and provides a benchmark for comparing them with the academic prototype developed in this project.

Most modern chargers available in the market employ Switch Mode Power Supply (SMPS) technology. Companies like Apple (USB-C Power Adapters), Samsung (Adaptive Fast Charging), Xiaomi (Quick Charge and HyperCharge), and Anker (PowerIQ) use SMPS-based designs because they offer high efficiency, reduced heat dissipation, smaller size, and higher current delivery. These chargers can support fast-charging protocols and power levels ranging from 15 W to over 100 W. However, SMPS chargers rely on complex switching circuits, feedback control, and high-frequency components, making them difficult to analyze and implement at a beginner or laboratory level.

A survey of commercially available adapters shows that standard output ratings include 5 V, 9 V, and 12 V, with current capacities from 1 A to several amperes, depending on the charging standard. Commercial chargers also integrate advanced protection mechanisms such as over-current protection, short-circuit protection, thermal shutdown, surge protection, and voltage regulation, ensuring user and device safety. Inspired by these safety practices, the proposed charger incorporates fuse protection at the output stage to safeguard the connected device during abnormal conditions.

Although chargers from reputed brands are highly efficient and compact, they are typically sealed, non-repairable, and use specialized components, making them unsuitable for academic experimentation or servicing. In contrast, the charger developed in this project uses readily available, low-cost components such as a transformer, rectifier diodes, filter capacitor, and linear regulator. This makes the design easy to understand, assemble, test, and repair in an educational laboratory environment. Hence, while commercial chargers focus on performance and miniaturization, the proposed design prioritizes learning, transparency, and practical demonstration, making it ideal for academic and training purposes.

1.5 Problem statement

Direct AC supply cannot be used for charging because devices require smooth and stable DC voltage. Hence, a regulated power conversion system is required to deliver constant DC output with minimal ripple under input and load conditions. The designed circuit successfully charged a mobile phone with proper regulation and fuse protection, while remaining simple, economical, and reliable.

1.6 Organization of the report

Chapter 1 – Introduction:
This chapter introduces the background and motivation for the project. It presents the objectives, problem statement, and provides an overview of the regulated power supply concept used for mobile charging.

Chapter 2 – Electronic System Design:
This chapter presents the overall design of the regulated charger. It includes alternative solutions, selection of the proposed method, and explanation of the functional block diagram.

Chapter 3 – Implementation Details:
This chapter describes the system specifications, circuit realization, and design calculations. The operational flow of the charger is also discussed.

Chapter 4 – Results and Discussions:
This chapter provides simulation results and hardware observations. The performance of the system is analyzed and verified.

Chapter 5 – Conclusion and Reflections:
This chapter summarizes the outcomes of the project and presents future improvements and possible applications.

Chapter 2

Electronic System design

A regulated charger is realized by interconnecting several functional blocks, where each block contributes to improving the quality of the output supply. The objective of the system is to convert the available AC mains power into a stable and reliable DC voltage suitable for charging mobile devices. The design follows a staged approach in which voltage transformation, rectification, filtering, and regulation are carried out sequentially.

The overall performance of the charger depends on the proper coordination between these stages. The transformer ensures electrical isolation and reduces the voltage to a safer level. The rectifier converts alternating current into pulsating DC, which still contains variations. The filter capacitor reduces these fluctuations by storing energy during peak levels and releasing it during drops. Finally, the regulator maintains a constant output voltage even when there are disturbances in the input supply or changes in the load current. Through this structured arrangement, the system achieves safety, efficiency, and dependable operation.

2.1 Alternative Solutions

Different techniques can be employed to achieve voltage regulation in DC power supplies. One of the simplest approaches is the Zener diode regulator. In this method, the Zener diode operates in its breakdown region and maintains a nearly constant voltage across the load. The circuit is easy to build, inexpensive, and requires very few components. Because of these advantages, Zener regulation is often used in small signal or low current applications.

However, Zener-based regulators suffer from certain limitations. They are less efficient, offer poor regulation for large load variations, and may experience higher power dissipation at increased currents. Thermal stability also becomes a concern when the load requirement rises. For applications like mobile charging, where current demand can approach higher values, Zener regulation may not provide satisfactory performance.

Considering these factors, integrated linear regulators such as the 7809 are preferred in the proposed system. They provide better load regulation, built-in protection features, and improved reliability while still maintaining design simplicity.

2.2 Functional block diagram

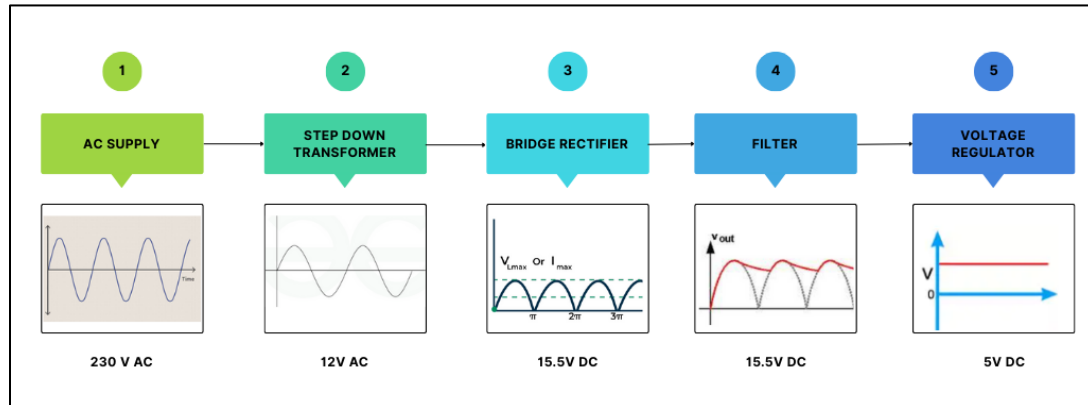


Fig 2.1 Functional Block Diagram

The block diagram represents the stages involved in converting AC mains power into a regulated DC output. Initially, the step-down transformer reduces the high input voltage to a lower AC level suitable for the circuit. The bridge rectifier then converts this AC into pulsating DC. A filter capacitor smooths the waveform by reducing ripple and fluctuations. Finally, the voltage regulator maintains a constant output voltage independent of variations in input or load. This sequential processing ensures that the power delivered to the connected device is safe, stable, and reliable for continuous operation. Each block plays a crucial role in improving the quality of the supply and protecting sensitive electronic components.

Chapter 3

Implementation details

The implementation of the 9 V USB mobile charger was carried out in a systematic manner, beginning with circuit design and component selection based on required electrical specifications. A step-down transformer, bridge rectifier, filter capacitor, and IC 7809 voltage regulator were interconnected to achieve AC–DC conversion and voltage regulation. The circuit was first verified through simulation to ensure correct operation and expected voltage levels at each stage. After successful simulation, the hardware prototype was assembled on a perfboard using readily available components, and measurements were taken to validate proper functioning. An output fuse was incorporated to provide protection against overcurrent and short-circuit conditions, enhancing the overall safety of the charger.

3.1 Bill of Materials

Sl. No.	Component	Specification	Purpose
1	Step-down Transformer	230 V AC to 12 V AC, 1 A	Reduces mains voltage
2	Diode	1N4007	Bridge rectification
3	Voltage Regulator IC	7809	Maintains constant 9 V
4	Electrolytic Capacitor	1000 μ F / 25 V	Ripple filtering
5	Ceramic Capacitor	0.01 μ F	Noise bypass / stability
6	Heat sink	Suitable for 7809	Dissipates regulator heat
7	USB Connector	As required	Connects phone/load
8	Breadboard / PCB	General purpose	Mounting circuit
9	Connecting wires	–	Electrical connections
10	Fuse (recommended)	250 mA – 500 mA	Short circuit protection

Table 3.1 Bill Of Materials of the Materials

3.2 Specifications and System Design

1. Input Specifications

- Input Voltage: 230 V AC $\pm$ 10%, 50 Hz
- Input Type: Single-phase mains supply
- Isolation provided using step-down transformer.

2. Transformer Specifications

- Type: Step-down iron core transformer
- Primary: 230 V AC
- Secondary: 12 V AC RMS

$$V_{in(min)} \approx V_{out} + 2V$$

$$V_{in(min)} = 9 + 2 = 11V$$

(Considering diode drops and ripple, a 12 V AC transformer is selected.)

- Rated Secondary Current: 1 A
- VA Rating: 12 VA minimum

3. Rectifier Stage Specifications

- Rectifier Type: Full-wave rectifier
- Diodes Used: 1N4007 $\times$ 4 (bridge configuration)
- Maximum Forward Current: 1 A (data sheet)
- Peak Inverse Voltage (PIV) Required: ≈ 34 V

$$V_{peak} = 12 \times 1.414 = 17V$$

$$PIV \approx 2 \times 17 = 34V$$

In a bridge rectifier, each diode must withstand approximately the peak voltage.

- 1N4007 PIV Rating: 1000 V (high safety margin) (Data sheet)

4. Rectified DC Voltage

- Secondary RMS voltage:

$$V_{rms} = 12V$$

- Peak voltage:

$$V_{peak} = 12 \times 1.414 = 16.97V$$

- After diode drops (≈ 1.4 V for bridge):

$$V_{DC} \approx 15.5V$$

5. Filter Capacitors

- C1 (Main Filter): 1000 μ F / 25 V
Let load current = 0.5 A

$$C = \text{Load current} / (2 \times \text{frequency})$$

$$C = 0.5 / (2 \times 100)$$

- Function: Reduces ripple from rectifier output
- Optional Output Capacitor: 0.01 μ F near regulator for stability

6. Voltage Regulator (7809)

- Nominal Output Voltage: 9 V DC
- Maximum Output Current: up to 1 A
- Dropout Voltage: approximately 2 V
- Internal Features: Thermal shutdown, short-circuit protection
- Recommended: Heat sink for continuous load operation

7. Power Dissipation in Regulator

- Power dissipated is 5W
- Input to regulator ≈ 14 V,

$$P = (V_{in} - V_{out}) \times I$$

$$P = (14 - 9) \times 1$$

$$P = 5W$$

8. Output Specifications

- Output Voltage: 9 V DC regulated
- Output Current Capacity: up to 1 A
- Ripple at Output: Very low (few millivolts due to regulation)
- Line and Load Regulation: Good for small variations

9. Fuse Selection and Calculation

- Rated output current of supply:

$$I_{rated} = 1A$$

- For protection, the fuse rating is generally chosen close to the maximum safe operating current so that it interrupts the circuit during overload or short-circuit conditions.

$$I_{fuse} \approx I_{rated}$$

$$I_{fuse} = 1A$$

- Selected fuse rating: 1 A
-

3.3 Flowchart

The flowchart illustrates the sequence of operations in the charger. The AC input is stepped down, rectified, filtered, and regulated to obtain stable DC output. A decision stage represents the fuse protection, which disconnects the load if excessive current occurs; otherwise, power is delivered safely to the mobile device.

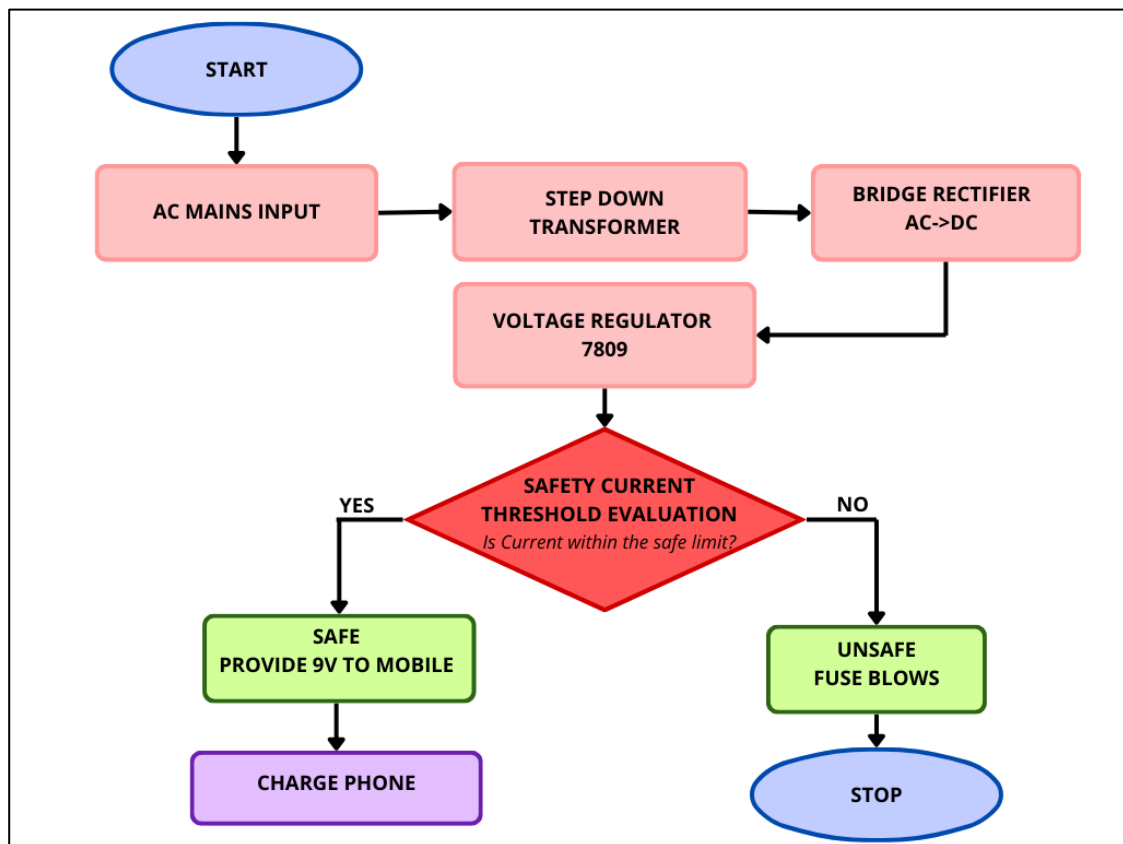


Fig 3.1 Flowchart

Chapter 4

Results and discussions

The simulation shows the conversion of AC input into DC through the transformer, bridge rectifier, filter capacitor, and 7809 regulator. The oscilloscope waveforms indicate ripple after rectification and filtering, while the final output remains nearly constant due to voltage regulation. This confirms that each stage of the circuit is functioning correctly and contributes to improving the quality and stability of the supply.

4.1 Simulation Analysis

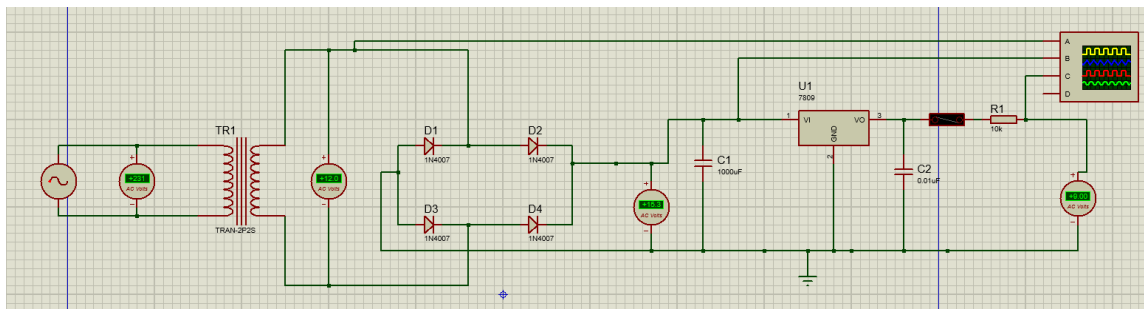


Fig 4.1 Simulation in Proteus

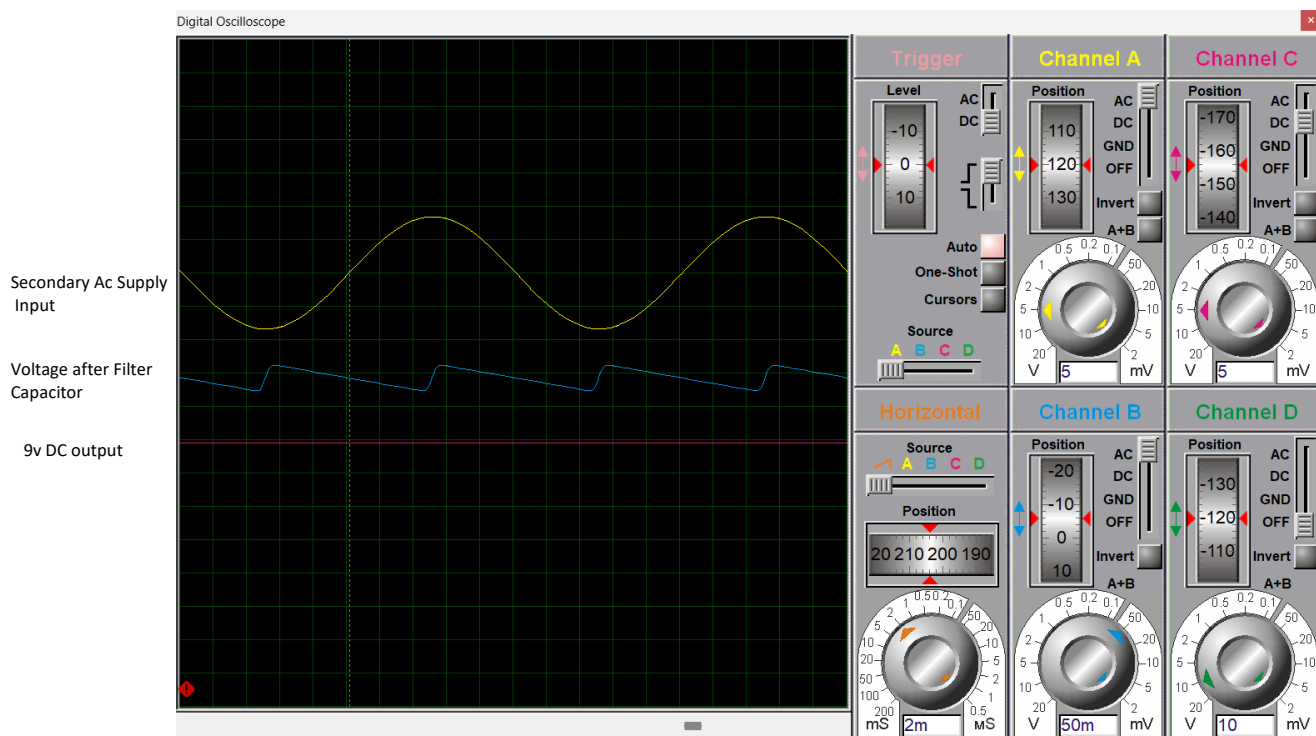


Fig 4.2 Waveforms in Oscilloscope

The simulation shows the conversion of AC input into DC through the transformer, bridge rectifier, filter capacitor, and 7809 regulator. The oscilloscope waveforms indicate ripple after rectification and filtering, while the final output remains nearly constant due to voltage regulation. This confirms that each stage of the circuit is functioning correctly and contributes to improving the quality and stability of the supply.

4.2 Hardware Realization

After verifying the circuit operation through simulation, the hardware prototype of the regulated charger was assembled using the selected components. Initially, voltage levels at different stages were measured using a digital multimeter. The transformer secondary provided the expected reduced AC voltage. After rectification, pulsating DC was observed, and once the filter capacitor was connected, the output became significantly smoother. The input to the 7809 regulator remained well above the minimum requirement for proper regulation, and the series output fuse was included to ensure protection against excessive current.

To validate real-time performance, a mobile phone was connected to the output of the charger. The device successfully detected the supply and indicated charging. The charging process remained stable without interruptions, confirming that the regulator maintained a constant output voltage under load conditions. In the event of an abnormal condition, the fuse would isolate the load, thereby improving safety. No abnormal heating or voltage drop was observed during short-duration testing, although the regulator became moderately warm, indicating normal power dissipation in linear regulation. Fig 4.3 and 4.4 shows the working of the mobile charger.

This practical experiment demonstrated that the designed circuit is not only theoretically correct but also capable of supporting real charging applications with additional protection features. The successful charging of a mobile device validated the reliability, safety, and usability of the system.

4.3 Result at Each Stage

Stage of Circuit	Expected Value	Measured Value
Transformer Secondary (AC)	12 V AC	11.8 – 12.4 V AC
After Rectifier (DC, no filter)	~15–17 V peak	~15.5 V DC
After Filter Capacitor	Smoothed DC	~14.2 V DC
Regulator Input	> 11 V	~14 V
Final Output (7809)	9 V	8.95 – 9.05 V

Table 4.1 Result at each stage

4.4 Image of hardware

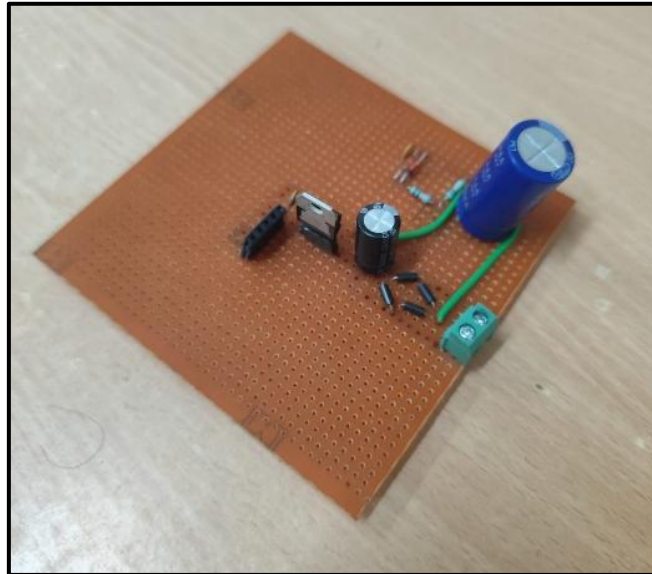


Fig 4.3 Working Circuit On Perfboard

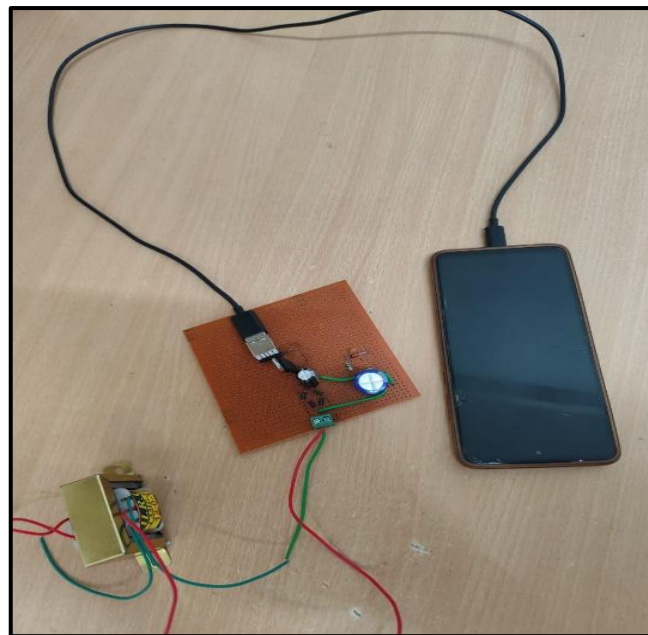


Fig 4.4 Phone Charging

Chapter 5

Conclusion and Reflections

The project successfully implemented a regulated 9 V USB mobile charger using basic electronic components. Stable DC output was achieved through rectification, filtering, and voltage regulation, with fuse protection ensuring safe operation. Simulation and hardware results validated the design. The project strengthened practical understanding of power supply design and testing.

5.1 Conclusion

The mini project successfully demonstrated the design and implementation of a regulated 9 V DC mobile charger using a transformer, bridge rectifier, filter capacitor, voltage regulator, and an output fuse for enhanced safety. The system effectively converted AC mains supply into a stable DC output suitable for practical applications. Both simulation and hardware testing confirmed that the circuit performs according to theoretical expectations. The rectifier converted AC into pulsating DC, the filter significantly reduced ripple, and the regulator maintained a constant voltage despite variations in operating conditions. The inclusion of the fuse ensured protection against abnormal current levels. Most importantly, the constructed prototype was able to charge a mobile phone successfully, thereby validating the usefulness, safety, and reliability of the design in real-time usage.

Through this project, we gained hands-on experience in circuit assembly, measurement techniques, protection methods, and troubleshooting. We also developed a deeper understanding of voltage regulation, power dissipation, and the importance of correct component selection. The work bridges academic concepts with practical implementation and provides a strong foundation for further exploration in power electronics.

5.2 Future Scope

While the present charger meets the fundamental objective of providing regulated DC output, there are several opportunities for enhancement. Efficiency can be improved by adopting modern switching regulator techniques that reduce heat generation and power loss. Advanced protection circuits such as electronic current limiting, short-circuit detection, and reverse polarity safeguards can be incorporated to further increase durability and user safety. Future versions may include smart charging features, automatic current adjustment, fast-charging compatibility, and digital monitoring of voltage and current parameters. Integration of standard USB interfaces, compact PCB layouts, and better thermal design would make the system more suitable for everyday consumer applications. Improved heat management can enable the charger to deliver higher currents without affecting reliability.

5.2.1 Application in the societal context

The developed charger demonstrates that reliable and safe power supply solutions can be built using simple and affordable components. Such designs are particularly valuable in educational environments, where students gain practical exposure to electronics through hands-on experimentation and testing.

In addition, low-cost regulated chargers can support communities where access to commercial products may be limited or expensive. Understanding how to build, maintain, and protect such systems promotes self-reliance, encourages repair culture, and contributes to reducing electronic waste. The inclusion of safety features like fuse protection further enhances usability in real-world conditions.

By combining affordability, simplicity, protection, and functional performance, the project contributes positively to both technical learning and societal development.

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